

Variation of params : Find a particular solution for

$$y'' + 4y = \cos(9x) \leftarrow f(x) \neq 0$$

find y_c : $y'' + 4y = 0 \rightarrow r^2 + 4 = 0$
 $r = \pm 2i$ (mult. 1)

so general solution $y_c = c_1 \underline{\cos(2x)} + c_2 \underline{\sin(2x)}$
 $y_1 = \cos(2x) \quad y_2 = \sin(2x)$

Variation of params is:

find u_1, u_2 (functions) such that

$$y = u_1 y_1 + u_2 y_2$$

method leads to

$$\begin{cases} u_1' y_1 + u_2' y_2 = 0 \\ u_1' y_1' + u_2' y_2' = f = \cos(9x) \end{cases}$$
$$\begin{cases} u_1' \cos(2x) + u_2' \sin(2x) = 0 \\ u_1' \cdot (-2\sin(2x)) + u_2' \cdot (2\cos(2x)) = \cos(9x) \end{cases}$$

which (always) has solutions

$$\begin{cases} u_1' = \frac{-y_2 \cdot f}{W(y_1, y_2)} \\ u_2' = \frac{+y_1 \cdot f}{W(y_1, y_2)} \end{cases}$$

$$\begin{aligned} W(y_1, y_2) &= \begin{vmatrix} \cos(2x) & \sin(2x) \\ -2\sin(2x) & 2\cos(2x) \end{vmatrix} \\ &= 2\cos^2(2x) + 2\sin^2(2x) \\ &= 2 \end{aligned}$$

$$u_1' = \frac{-\sin(2x) \cdot \cos(9x)}{2}$$

$$u_2' = \frac{\cos(2x) \cos(9x)}{2}$$

hint: trig identities

$$\sin(a) \cos(b) = \frac{1}{2} (\sin(a-b) + \sin(a+b))$$

$$\cos(a) \cos(b) = \frac{1}{2} (\cos(a-b) + \cos(a+b))$$

$$\begin{aligned} u_1' &= \frac{-\sin(2x) \cdot \cos(9x)}{2} = -\frac{1}{4} (\sin((2-9)x) + \sin((9+2)x)) \\ &= -\frac{1}{4} (-\sin(7x) + \sin(11x)) \end{aligned}$$

$$u_1 = \int u_1' dx = +\frac{1}{4} \left(-\frac{1}{7} \cos(7x) + \frac{1}{11} \cos(11x) \right) + C_1$$

$$u_2' = \frac{\cos(2x) \cos(9x)}{2} = \frac{1}{4} (\cos(7x) + \cos(11x))$$

$$u_2 = \int u_2' dx = \frac{1}{4} \left(\frac{1}{7} \sin(7x) + \frac{1}{11} \sin(11x) \right) + C_2$$

$$u_1 y_1 + u_2 y_2 =$$

$$C_1 \cos(2x) + C_2 \sin(2x)$$

$$+ \frac{-1}{28} \cos(7x) \cos(2x) + \frac{1}{44} \cos(11x) \cos(2x)$$

$$+ \frac{1}{28} \sin(7x) \sin(2x) + \frac{1}{44} \sin(11x) \sin(2x)$$

$$\text{hint: } \cos(a) \cos(b) + \sin(a) \sin(b) = \cos(a-b)$$

$$\cos(a) \cos(b) - \sin(a) \sin(b) = \cos(a+b)$$

$$u_1 y_1 + u_2 y_2 = c_1 \cos(2x) + c_2 \sin(2x) - \frac{1}{28} \cos(9x) + \frac{1}{44} \cos(9x)$$

$$= \underbrace{c_1 \cos(2x) + c_2 \sin(2x)}_{\text{this is } y_c} - \underbrace{\frac{1}{77} \cos(9x)}_{\text{so this is } y_p}$$

this is y_c , so this is y_p

Much easier way: Since $\cos(9x)$ is one of our "known cases", we can use Undetermined coeffs:

$$\begin{aligned} y'' + 4y &= \cos(9x) \\ y_c &= c_1 \cos(2x) + c_2 \sin(2x) \\ y_p &= A \cos(9x) + B \sin(9x) \\ y_p' &= -9A \sin(9x) + 9B \cos(9x) \\ y_p'' &= -81A \cos(9x) - 81B \sin(9x) \end{aligned}$$

$$\begin{aligned} y_p'' + 4y_p &= (4-81)A \cos(9x) + (4-81)B \sin(9x) \\ &= \cos(9x) \end{aligned}$$

$$A = -\frac{1}{77}$$

$$B = 0,$$

$$\text{so } y_p = -\frac{1}{77} \cos(9x)$$